Personalized Learning and Career Path Analyzer using Agentic AI

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***Abstract*—The AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) is an intelligent web-based platform designed to provide personalized career guidance and learning recommendations for students. Many students face difficulties in selecting the right career path due to a lack of proper guidance, skill assessment, and awareness of employment and higher education opportunities. This system addresses these challenges by collecting student information such as academic performance, skills, interests, and career goals, and utilizing Artificial Intelligence (Groq Llama 3.3 API) to generate personalized career recommendations, identify skill gaps, and suggest learning roadmaps. The platform supports three user roles—Student, Faculty, and Admin—allowing students to manage their profiles and receive AI-driven guidance, faculty members to monitor student progress, and administrators to manage users and system data efficiently. Developed using React.js for the frontend, Spring Boot for the backend, PostgreSQL for database management, and RESTful APIs for communication, the system offers a centralized solution for career planning. By integrating AI with educational data, the platform enhances decision-making, promotes continuous skill development, and helps students prepare for private jobs, government jobs, and higher education opportunities, ultimately improving their employability and career readiness.**

***Index Terms*—Personalized Learning, Career Path Analyzer, Agentic AI, Llama 3.3, Full-Stack Web Application, Decision Support System.**

I. Introduction

In today's competitive world, students often struggle to choose the right career path due to a lack of personalized guidance, skill assessment, and awareness of available opportunities. Traditional career counseling methods are usually generic, time-consuming, and unable to provide recommendations based on an individual student's academic performance, technical skills, interests, and career aspirations. As a result, many students face uncertainty while preparing for jobs, higher education, or competitive examinations.

The AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) is developed to address these challenges by providing an intelligent and personalized career guidance platform. The system utilizes Artificial Intelligence (Groq Llama 3.3 API) to analyze a student's profile, including academic details, skills, interests, and career goals, and generates customized career recommendations, identifies skill gaps, and suggests learning roadmaps. It also provides information about private sector jobs, government job opportunities, and higher education options based on the student's profile.

The platform supports three types of users: Students, Faculty, and Administrators. Students can register, manage their profiles, complete skill assessments, and receive AI-powered recommendations. Faculty members can monitor student progress and provide guidance, while administrators manage users, departments, and system operations. The application is developed using React.js for the frontend, Spring Boot for the backend, PostgreSQL as the database, and RESTful APIs for seamless communication between system components.

By integrating artificial intelligence with modern web technologies, the AI-PLCPS platform provides an efficient, scalable, and user-friendly solution for personalized career planning. It helps students make informed career decisions, improve their technical skills, and prepare effectively for employment and higher education, thereby enhancing their overall career readiness and employability.

II. Literature Review

Career guidance has become an important area of research due to the increasing demand for skilled professionals and the rapid growth of technology. Traditional career counseling methods often provide generalized suggestions and require significant human intervention, making it difficult to offer personalized guidance to every student. Researchers have therefore explored the use of Artificial Intelligence (AI), Machine Learning (ML), and web-based technologies to develop intelligent career guidance systems that provide customized recommendations based on students' academic performance, skills, interests, and career goals.

Several studies have demonstrated that AI-based recommendation systems can significantly improve career decision-making by analyzing user profiles and suggesting suitable career paths. Machine learning algorithms and Natural Language Processing (NLP) techniques have been widely used to process student data and generate personalized recommendations. Recent advancements in Large Language Models (LLMs), such as Llama and GPT-based models, have further enhanced the quality of career guidance by providing detailed explanations, learning roadmaps, and skill development suggestions in natural language.

Web-based educational platforms have also contributed to personalized learning by integrating skill assessment, learning resources, and progress tracking into a single platform. These systems help students identify their strengths and weaknesses while recommending appropriate courses and certifications. However, many existing systems focus only on either learning recommendations or job suggestions and often lack comprehensive career planning features that include higher education opportunities, government jobs, and private sector employment.

The proposed AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) addresses these limitations by integrating Artificial Intelligence with a modern full-stack web application. The system uses React.js for the user interface, Spring Boot for backend services, PostgreSQL for secure data management, and the Groq Llama 3.3 API to generate intelligent career recommendations. Unlike traditional systems, AI-PLCPS provides a complete solution by combining student profile management, AI-powered career recommendations, skill gap analysis, personalized learning roadmaps, private and government job suggestions, and higher education guidance within a single platform. This integrated approach enables students to make informed career decisions and improve their employability through personalized and data-driven recommendations.

III. Research Gap and Problem Statement

A. Research Gap

Although several career guidance and learning management systems have been developed, most existing solutions have significant limitations. Many systems provide only general career suggestions without considering a student's complete academic profile, technical skills, interests, and career goals. Existing platforms often focus on a single aspect, such as job recommendations, online learning, or skill assessment, rather than offering a comprehensive career planning solution. Additionally, traditional systems rely heavily on manual counseling, which can be time-consuming, expensive, and unable to provide personalized guidance to every student. Most available platforms also lack integration with modern Artificial Intelligence technologies capable of generating detailed career recommendations, skill gap analysis, and personalized learning roadmaps. Furthermore, they rarely include guidance for multiple career options such as private sector jobs, government employment, and higher education within a single application.

The proposed AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) addresses these research gaps by integrating Artificial Intelligence (Groq Llama 3.3 API) with a full-stack web application to provide intelligent, personalized, and real-time career guidance. The system combines student profile management, AI-driven career recommendations, skill gap identification, customized learning roadmaps, private and government job suggestions, and higher education guidance in one unified platform. It also supports multiple user roles—Student, Faculty, and Admin—making it a comprehensive solution for educational institutions. This integrated and AI-powered approach overcomes the limitations of existing systems and enhances students' career planning, skill development, and overall employability.

B. Problem Statement

Students often face difficulty in selecting suitable career paths due to a lack of personalized guidance, inadequate awareness of industry requirements, and limited access to reliable career counseling. Traditional career guidance methods are mostly manual, time-consuming, and unable to analyze a student's academic performance, technical skills, interests, and career aspirations comprehensively. As a result, many students struggle to identify skill gaps, choose appropriate learning resources, and explore relevant opportunities in private companies, government organizations, or higher education. Existing career guidance systems generally focus on a single aspect, such as job recommendations or online learning, and fail to provide an integrated solution that supports complete career planning. Therefore, there is a need for an intelligent, AI-powered platform that can analyze individual student profiles, generate personalized career recommendations, identify skill gaps, suggest learning roadmaps, and provide guidance for employment and higher education through a single, user-friendly system. The proposed AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) addresses this challenge by leveraging Artificial Intelligence, modern web technologies, and personalized recommendation techniques to assist students in making informed career decisions and improving their employability.

IV. Proposed Methodology

The proposed AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) is designed using a three-tier architecture consisting of a React.js frontend, a Spring Boot backend, and a MySQL database, with the Groq Large Language Model (LLM) integrated to provide AI-based career recommendations. The system begins by allowing users to select their role (Student, Faculty, or Admin) and log in through a secure authentication module. Student profile details such as academic performance, technical skills, career interests, and learning goals are stored in the database.

The backend retrieves this information and constructs a structured prompt, which is sent to the Groq AI API for analysis. Based on the student's profile, the AI generates personalized career recommendations, identifies skill gaps, suggests learning roadmaps, recommends private and government job opportunities, and provides higher education guidance. Faculty members can monitor student progress and performance, while administrators manage users, departments, and system activities. The system follows RESTful API communication between the frontend and backend, ensuring efficient data exchange and real-time responses.

A. Methodology Flow

The workflow of the methodology is systematically structured as follows:

1) Role Selection (Student / Faculty / Admin Login)
2) Authentication & Authorization
3) Student Profile Management
4) Data Stored in MySQL Database
5) Spring Boot Backend Retrieves Profile
6) Prompt Generation
7) Groq AI (LLM) Analysis
8) Personalized Career Recommendation (Includes Recommended Career, Skill Gap Analysis, Learning Roadmap, Private/Govt Job Suggestions, Higher Studies Guidance)
9) Results Displayed on React Dashboard

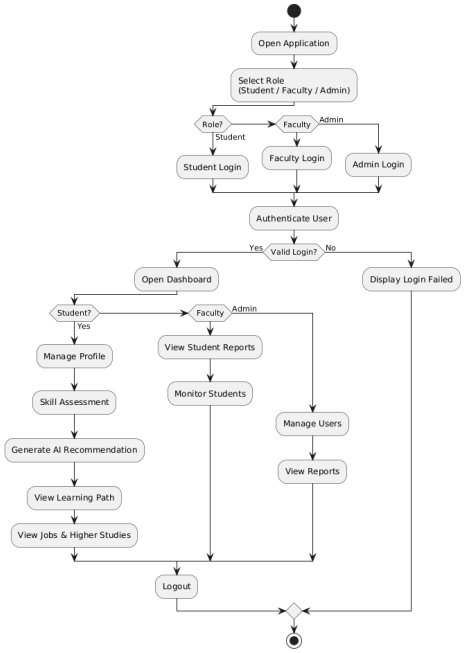


Fig. 1. Workflow of the Proposed AI-PLCPS Methodology.

This methodology ensures a complete, secure, and AI-driven career guidance process that combines academic data, skill assessment, and intelligent recommendations to support students in achieving their career goals.

V. System Architecture

The proposed AI-Based Personalized Learning and Career Prediction System follows a three-tier architecture consisting of the Presentation Layer (Frontend), Application Layer (Backend), and Database Layer. This architecture provides a scalable, secure, and modular platform for managing student information, generating AI-based career recommendations, and supporting multiple user roles such as Students, Faculty, and Administrators.

The Presentation Layer is developed using React.js with Material-UI to provide an interactive and responsive user interface. It allows users to register, log in based on their roles, manage profiles, complete skill assessments, explore learning paths, view career recommendations, and access government jobs, private jobs, and higher studies information. The frontend communicates with the backend through secure REST APIs.

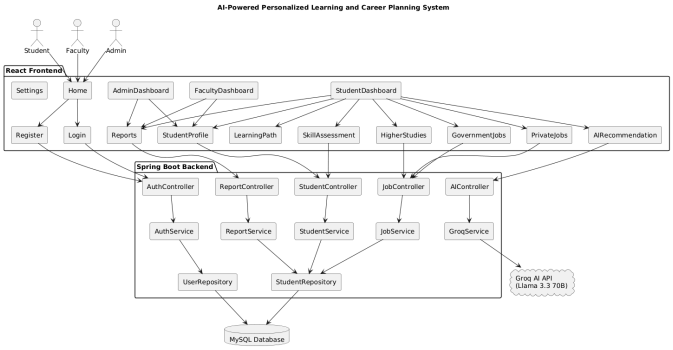


Fig. 2. Proposed Architecture of the AI-PLCPS Framework.

The Application Layer is built using Spring Boot, which handles all business logic and request processing. It manages authentication and authorization, student profile management, skill assessment processing, AI recommendation generation, report management, and role-based access control. For career prediction, the backend integrates with the Groq AI API (Llama 3.3 70B Versatile model) to analyze student academic performance, technical skills, career interests, and learning goals, producing personalized career recommendations and learning roadmaps.

The Database Layer uses MySQL to securely store user credentials, student profiles, assessment results, learning progress, reports, and recommendation history. Spring Data JPA and Hibernate provide efficient interaction between the application and the database, ensuring reliable data persistence and retrieval.

The overall workflow begins when a user selects a role (Student, Faculty, or Admin) and logs into the system. Students update their academic and skill information, which is stored in the database. The backend retrieves this information, constructs an AI prompt, and sends it to the Groq AI model. The AI analyzes the student's profile and returns recommendations including suitable career paths, skill gaps, learning roadmaps, private companies, government opportunities, and higher education suggestions. The generated results are displayed on the dashboard, enabling users to make informed career decisions while faculty and administrators monitor student progress through dedicated dashboards.

VI. Implementation and Experimental Evaluation

A. Implementation

The implementation of the AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) was carried out using a full-stack web development approach. The frontend was developed using React.js and Material-UI, providing a responsive and user-friendly interface for Students, Faculty, and Administrators. The backend was implemented using Spring Boot, which exposes RESTful APIs to handle user authentication, profile management, skill assessment, report generation, and communication with the AI model. MySQL

was used as the relational database to securely store user credentials, student profiles, academic details, assessment records, and recommendation history. Spring Data JPA and Hibernate were utilized for efficient database operations.

The system begins by allowing users to select their role and log in securely. Students can create and update their profiles by entering academic information, technical skills, career interests, and career goals. This information is stored in the database and retrieved by the backend whenever an AI recommendation is requested. The backend constructs a detailed prompt using the student's profile and sends it to the Groq AI API (Llama 3.3 70B Versatile model). The AI analyzes the provided information and generates personalized career recommendations, identifies skill gaps, suggests learning roadmaps, recommends private companies and government job opportunities, and provides higher education guidance. The generated recommendations are then displayed on the React dashboard in an easy-to-understand format.

B. Experimental Evaluation

The experimental evaluation of the AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) was conducted to assess its functionality, performance, and effectiveness in providing personalized career guidance. The system was tested using different student profiles with varying academic records, technical skills, career interests, and career goals. Each module, including user authentication, profile management, skill assessment, AI recommendation generation, learning path generation, and report management, was evaluated to ensure correct functionality and seamless integration.

The AI recommendation module was tested by providing different combinations of student data to the Groq Llama 3.3 API. The generated recommendations were evaluated based on their relevance, completeness, and personalization. The system successfully identified suitable career paths, highlighted skill gaps, suggested appropriate learning roadmaps, and recommended private companies, government job opportunities, and higher education options according to each student's profile. The response time for generating recommendations was observed to be within a few seconds under normal network conditions, providing a smooth user experience.

C. Experimental Results

Performance testing demonstrated that the React frontend and Spring Boot backend communicated efficiently through RESTful APIs, while MySQL handled data storage and retrieval with minimal latency.

The experimental evaluation demonstrates that the proposed AI-PLCPS effectively integrates Artificial Intelligence with modern web technologies to provide accurate, personalized, and efficient career guidance. The system improves students' decision-making capabilities by offering data-driven recommendations and supports educational institutions in delivering comprehensive career planning services.

TABLE I
FUNCTIONAL MODULE EVALUATION RESULTS

Module	Evaluation Result
User Registration & Login	Successfully authenticated users based on role
Student Profile Management	Accurate storage and retrieval of student data
Skill Assessment	Successfully recorded and displayed skill information
AI Career Recommendation	Generated personalized career guidance based on student profile
Learning Path Generation	Recommended relevant technologies and learning roadmap
Job Recommendation	Suggested suitable private and government job opportunities
Higher Studies Guidance	Recommended appropriate higher education options
Database Performance	Fast and reliable data storage and retrieval
Overall System Performance	Stable, responsive, and user-friendly

VII. RESULTS AND DISCUSSION

The implementation of the AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) successfully achieved the objectives of providing personalized career guidance through Artificial Intelligence. The developed system integrates React.js, Spring Boot, MySQL, and the Groq Llama 3.3 API to deliver a comprehensive platform for students, faculty, and administrators.

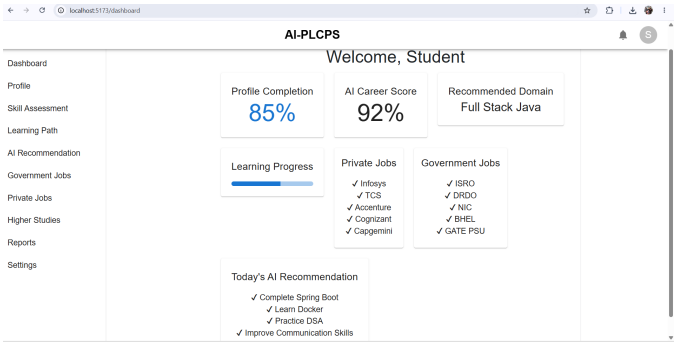


Fig. 3. Dashboard Interface Displaying AI Recommendations.

Students were able to register, manage their profiles, complete skill assessments, and receive AI-generated career recommendations based on their academic performance, technical skills, interests, and career goals. The AI module effectively identified skill gaps, suggested suitable learning roadmaps, recommended private companies, government job opportunities, and higher education options tailored to each student's profile.

The experimental results demonstrated that the system provides accurate, relevant, and personalized recommendations while maintaining fast response times and efficient data processing. Role-based authentication ensured secure access for Students, Faculty, and Administrators, enabling each user to perform their respective tasks without unauthorized access. The React frontend provided a responsive and user-friendly

interface, while the Spring Boot backend efficiently handled API requests and database operations. Overall, the proposed system proved to be reliable, scalable, and effective in supporting career planning and decision-making.

VIII. Conclusion

The AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) successfully demonstrates the application of Artificial Intelligence in providing personalized career guidance and learning recommendations for students. By integrating React.js, Spring Boot, MySQL, and the Groq Llama 3.3 API, the system offers a comprehensive platform that analyzes students' academic performance, technical skills, interests, and career goals to generate customized career recommendations. The platform effectively identifies skill gaps, suggests personalized learning roadmaps, and provides guidance for private sector jobs, government employment, and higher education opportunities. The inclusion of role-based access for Students, Faculty, and Administrators further enhances system usability and management.

The developed system overcomes the limitations of traditional career counseling by delivering intelligent, real-time, and data-driven recommendations through an easy-to-use web application. Experimental evaluation confirmed that the platform performs efficiently, provides relevant recommendations, and ensures secure management of user information. Overall, the proposed AI-PLCPS serves as an effective decision-support tool that helps students make informed career choices, improve their skills, and enhance their employability.

IX. Future Work

The proposed AI-Powered Personalized Learning and Career Planning System (AI-PLCPS) provides an effective platform for career guidance; however, several enhancements can further improve its functionality and impact. Future work will focus on integrating advanced Artificial Intelligence and Machine Learning models to provide more accurate, adaptive, and personalized career recommendations based on students' learning behavior, academic progress, and industry trends. The system can also be connected to real-time job portals such as LinkedIn, Naukri, and Indeed to display current job openings, internship opportunities, and company-specific hiring requirements.

Further improvements may include implementing predictive analytics to estimate students' employability scores and suggest career paths based on future market demand. The platform can also support resume analysis, AI-based mock interviews, aptitude assessments, and coding challenges to improve placement readiness. Faculty and administrators can be provided with advanced analytics dashboards to monitor student progress, identify weak areas, and generate institution-wide placement reports. Additional features such as multilingual support, mobile application development, cloud deployment, chatbot assistance, and enhanced security mechanisms using JWT and OAuth authentication can make the system more scalable, secure, and accessible.

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